World Population Trend and CO_2 Emission

R for Data Science – Project I, MSc in Big Data Analytics, AUCA

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Abstract

This project studies the relationship between population and carbon dioxide (CO_2) emissions using two Kaggle datasets: the World Population Dataset (234 countries and territories, 1970–2022) and CO_2 Emissions Around the World (metric tons per capita, 1990–2019). In R, the data were explored and cleaned, the 2030 population of every country was projected with $N_t = Pe^{rt}$, the ten most populous countries were visualised, and correlations were computed among area, density, growth rate and world population share. The two datasets were merged by country name (191 countries) to relate 2022 population to 2019 emissions and to compare continents. There were no duplicates or missing values, and the outliers were genuine countries, so they were kept. World population is projected to grow from 7.97 billion in 2022 to 8.56 billion in 2030, with India overtaking China. Area and world population percentage are strongly rank-correlated (Spearman 0.84), but population size is not related to per-capita CO_2 (Pearson 0.014, Spearman -0.11, $R^2 = 0.002$). Summing 2019 emissions per capita, Asia ranks first (290.5), Europe second (246.7), North America third (96.9) and South America last (33.4). Because the emission variable is per capita, this sum depends on the number of countries per continent and should be interpreted with caution.

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1 Introduction

Population growth is often linked to environmental pressure, yet total population and emissions per person are different quantities. Knowing whether larger or faster-growing populations emit more CO_2 per person matters for climate and development policy. The objectives of this project were to (i) apply data management and exploratory analysis, (ii) create a new variable for the projected population, (iii) visualise population and emission trends for the most populous countries, and (iv) analyse correlations between population and emissions at country and continent level.

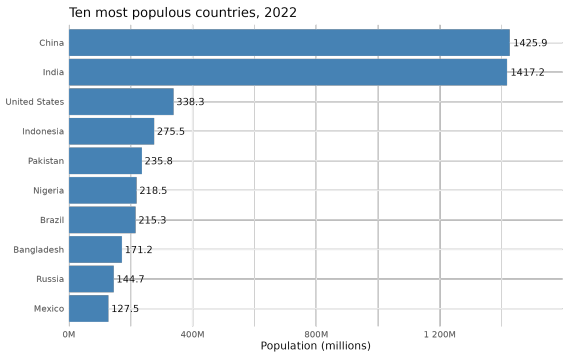
2 Methods

Datasets. The World Population Dataset has 234 rows and 17 variables: rank, country code (CCA3), country, capital, continent, population for 1970, 1980, 1990, 2000, 2010, 2015, 2020 and 2022, area (km^2), density, growth rate and world population percentage. The CO_2 dataset has 215 countries with yearly emissions in metric tons *per capita* for 1990–2019; the 2019 column was duplicated and one copy was removed. **Procedure.** All work was done in R (dplyr, tidyr, ggplot2) and knitted with R Markdown. We displayed variable names, the first five and last ten rows, data types and shape, counted duplicates and missing values, and used box plots and the 1.5×IQR rule for outliers. The growth rate in the data is a yearly multiplier (e.g. 1.0257), so the continuous rate is $r = \ln(\text{growth rate})$ and $N_{2030} = P_{2022}\, e^{8r}$. Country names in the two files were harmonised (29 names, e.g. "Russian Federation" → "Russia") and merged on country name; continents come from the population dataset. Correlations used Pearson and Spearman coefficients with heatmaps, and a regression of 2019 CO_2 on $\log_{10}$ of 2022 population.

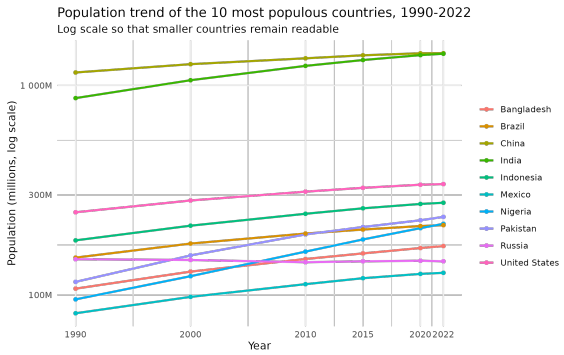
3 Results

Exploration (4.2). There were no duplicate rows and no missing values. Box plots flagged 25 outliers for 2022 population, 28 for area, 22 for density, 3 for growth rate and 25 for world population percentage. These are genuine countries (China, India, Russia, Macau, Monaco), so they were kept and log transformation and rank-based correlation limited their influence.

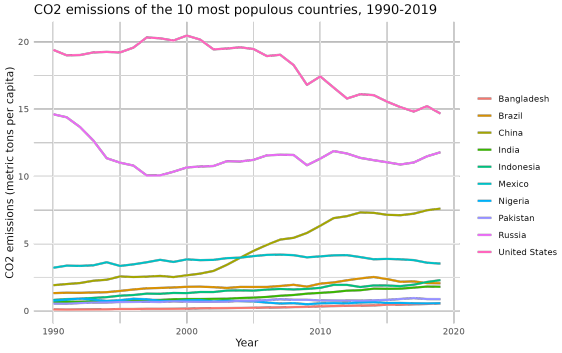
Projection (4.3). World population rises from 7.97 billion (2022) to 8.56 billion (2030). India reaches about 1.50 billion and overtakes China, whose growth rate is about 1.000 and whose population stays near 1.43 billion. Nigeria



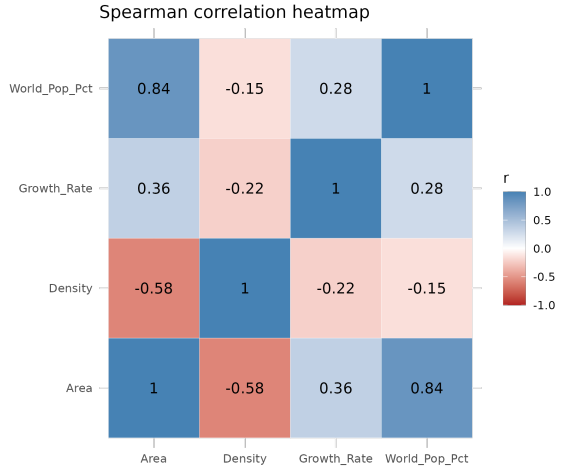
(a) Top 10 population, 2022



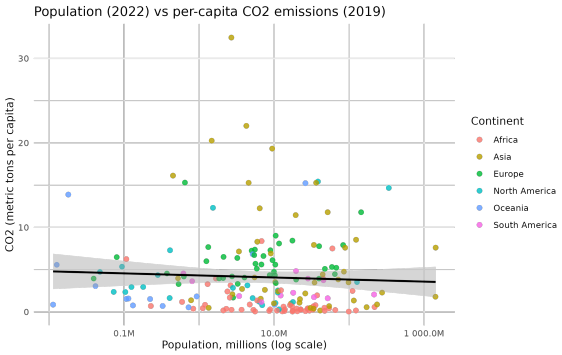
(b) Population trend 1990–2022



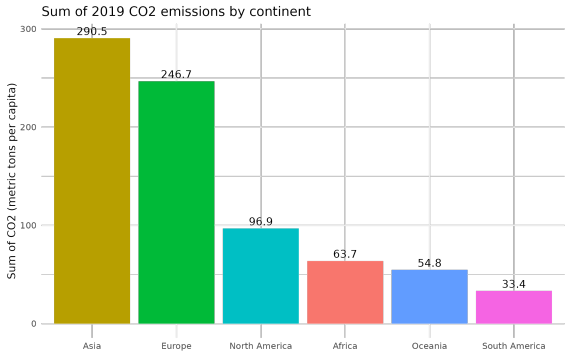
(c) CO_2 per capita 1990–2019



(d) Spearman correlation



(e) Population vs CO_2



(f) Summed 2019 CO_2 by continent

Figure 1: Main results: (a)–(c) the ten most populous countries; (d) correlation of area, density, growth rate and world population percentage; (e) 2022 population against 2019 per-capita CO_2; (f) continents.

(218.5M → 264.4M) and Pakistan (235.8M → 274.4M) grow fastest among the top ten, while Russia (144.7M → 141.6M) declines.

Top ten countries (4.4). China, India, the United States, Indonesia, Pakistan, Nigeria, Brazil, Bangladesh, Russia and Mexico (Fig. 1a) hold 57% of the world population. Since 1990, Nigeria (2.3×) and Pakistan (2.0×) grew fastest, and Russia (0.98×) is the only one that shrank (Fig. 1b). Per-capita CO_2 rose strongly in China (1.9 to 7.6 t), India (0.6 to 1.8 t) and Indonesia (0.8 to 2.3 t) and fell in the United States (19.4 to 14.7 t) and Russia (14.6 to 11.8 t) (Fig. 1c).

Correlations (4.5). Pearson coefficients were weak, except area against world population percentage (0.45). Spearman coefficients were stronger: area and world population percentage 0.84, area and density −0.58, area and growth rate 0.36 (Fig. 1d). After the merge, 191 countries had both 2022 population and 2019 CO_2. Population and per-capita CO_2 were not related: Pearson $r = 0.014$ ($p = 0.84$), Spearman $\rho = -0.11$ ($p = 0.13$), and the regression on log population has slope −0.24 ($p = 0.50$) with $R^2 = 0.002$ (Fig. 1e).

Continents (4.6). Summing 2019 per-capita emissions (Fig. 1f): (1) **Asia** is first with 290.5; (2) Europe is second with 246.7; (3) the third continent is **North America**, with a total of 96.9 metric tons per capita; (4) Africa 63.7 and Oceania 54.8 follow; (5) the last continent is **South America**, with a total of 33.4 metric tons per capita.

4 Discussion

The absence of a relationship between population size and per-capita CO_2 is expected: emissions per person depend mainly on income, energy mix and industrial structure, not on country size. China and India show the rapid rise of emissions with industrialisation, while the United States and Russia are high emitters whose per-capita values are slowly falling. The continent totals need care. Adding per-capita values counts a small island like China and gives larger sums to continents with many countries (Africa has 54, Asia 46). The average per country gives another order: Asia 6.3, Europe 5.9, North America 4.2, Oceania 3.9, South America 2.8 and Africa 1.2, so Africa becomes the lowest. The projection assumes a constant growth rate and ignores migration and shocks, so it is indicative only. The population file also has only eight census years, which limits trend detail.

5 Lessons Learned and Recommendations

Lessons. (i) Check units first: growth rate is a multiplier and CO_2 is per capita, and both change the correct method. (ii) Country names must be harmonised before merging, otherwise countries are lost silently. (iii) With strong outliers, Spearman correlation and log scales are more reliable than Pearson on raw values. **Recommendations.** Use total emissions (per-capita value $\times$ population) or population-weighted means for continent comparisons; add GDP, energy mix and urbanisation to explain emissions; use annual population series and scenario-based projections; and merge on ISO country codes rather than names.

6 Conclusion

The world population is projected to reach about 8.56 billion by 2030, with India overtaking China. Population size alone does not explain per-capita CO_2 emissions, and the ranking of continents depends on how emissions are aggregated: by simple sum, Asia, Europe and North America lead and South America is last. Reliable conclusions need correct units, careful merging and complementary economic variables.